NSD-ISS Stage Prediction with Calibrated Uncertainty: A Benchmarked Machine Learning Framework for Biological Staging in Parkinson's Disease

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The Neuronal alpha-Synuclein Disease Integrated Staging System (NSD-ISS) redefines Parkinson's disease (PD) through two biological anchors—neuronal alpha-synuclein (S, assessed by seed amplification assay) and dopaminergic dysfunction (D, assessed by DaT-SPECT)—yet no computational models exist to predict NSD-ISS stages from routinely collected clinical data. Using the Parkinson's Progression Markers Initiative (PPMI, n=2,201), we benchmarked eight machine-learning models (seven gradient-boosted and kernel-based tabular classifiers and one graph-based Multimodal Graph Attention Network) across four clinically-motivated target formulations. Calibrated uncertainty was quantified by cross-conformal prediction (CV+), a distribution-free procedure that returns a small set of plausible labels rather than a single prediction with a mathematical coverage guarantee, using the Least Ambiguous set-valued Classifier (LAC) nonconformity score. CatBoost achieved 0.951 balanced accuracy (AUC 0.979) for binary NSD-ISS classification, outperforming the Multimodal GAT (0.825) by 12.6 percentage points, consistent with the known dominance of tree ensembles on medium-sized tabular datasets. CV+ conformal prediction achieved $>$ 90% marginal coverage with near-singleton prediction sets (mean size 0.96–1.27 labels per patient). Removing DaT-SPECT features reduced binary AUC by 25.2%, while clinical features alone achieved AUC 0.900 for NSD-positive sub-staging—supporting a two-stage deployment pattern in which biological confirmation occurs once and subsequent clinical-only sub-staging supports sites lacking imaging access. External validation on BioFIND (n=118, PD-only cohort) revealed a training-label confound (binary balanced accuracy 0.516): PPMI's NSD-negative class mixes healthy controls with prodromal patients, causing the model to learn a healthy-vs-PD boundary. Retraining the binary classifier on PD-only subjects mitigates this confound (see §IV). Taken together, these findings establish the first computational benchmark for NSD-ISS biological staging with calibrated uncertainty quantification and a deployment-realistic assessment of label-structure effects.

Manuscript submitted April 2026. This work was supported by the University of North Dakota.

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Data used in the preparation of this article were obtained from the Parkinson's Progression Markers Initiative (PPMI) database (https://www.ppmi-info.org/access-data-specimens/download-data) and the Accelerating Medicines Partnership Parkinson's Disease (AMP-PD) platform (https://amp-pd.org).

I. Introduction

Parkinson's disease (PD) has traditionally been staged using clinical severity scales such as the Movement Disorder Society-sponsored Unified Parkinson's Disease Rating Scale (MDS-UPDRS) [18] and the MDS clinical diagnostic criteria [26]. These scales describe downstream motor manifestations of pathology that has typically been developing for years: pathological staging of Lewy pathology was proposed by Braak et al. [5], long before cerebrospinal-fluid alpha-synuclein seed amplification assays and dopamine-transporter (DaT) SPECT made in vivo biological staging operational at scale in cohorts such as the Parkinson's Progression Markers Initiative (PPMI) [22]. In January 2024, Simuni et al. [36] proposed the Neuronal alpha-Synuclein Disease Integrated Staging System (NSD-ISS), redefining PD through two biological anchors: neuronal alpha-synuclein (S, assessed by seed amplification assay, SAA) and dopaminergic dysfunction (D, assessed by DaT-SPECT imaging). The NSD-ISS assigns patients to stages 0 through 6, creating a biologically-grounded framework spanning preclinical through severe disability, and has been validated longitudinally: biological stage predicts time-to-clinical-milestones [37], staging replicates across cohorts including BioFIND [30], and 5-year stage transitions show consistent median times of 1.2 years (2B to 3) and 5.0 years (3 to 4) [38].

The NSD-ISS framework has also attracted principled critiques. Espay et al. [14] showed that dopaminergic medication status confounds several of the clinical sub-staging variables, leading to the recommendation that staging models be trained on cohorts with explicit medication-status stratification—a constraint we adopt in the present work and discuss further in §V.

TABLE I
FOUR NSD-ISS TARGET FORMULATIONS EVALUATED IN THIS STUDY.

Target	K	What it predicts
Binary	2	NSD+ (stages 1+) vs. NSD− (stage 0)
Three-class	3	Early (0–1) vs. mild (2B) vs. impaired (3–4)
Full ordinal	5	All stages: 0, 1, 2B, 3, 4
NSD-positive	4	NSD+ sub-staging: 1, 2B, 3, 4

Despite rapid clinical adoption, the NSD-ISS literature remains predominantly descriptive—no machine-learning model has been reported that predicts NSD-ISS biological stage from routinely-collected PPMI data with calibrated uncertainty, and the related PD-ML literature [1], [11], [28] has not targeted the NSD-ISS framework directly. Reporting guidance for such clinical prediction models is provided by the TRIPOD+AI statement [9]. The primary research question of this study is therefore: can a benchmarked machine-learning framework with calibrated uncertainty predict NSD-ISS biological stage from routinely-collected PPMI clinical and imaging data, including cross-cohort transportability and a clinical-only feature subset for sites lacking DaT-SPECT or SAA access? Five specific objectives operationalise this question:

1) Benchmark seven tabular and one graph-based model for NSD-ISS stage prediction across four target formulations using PPMI data (n=2,201, §III).
2) Compare tabular (gradient-boosted trees) versus graph-based (graph attention network (GAT) with patient-similarity graphs) approaches on an identical feature set.
3) Quantify the contribution of biological markers versus clinical-only features, using a feature-ablation protocol matched to site-level resource availability.
4) Evaluate model transportability through external validation on BioFIND (n=118, with NSD-ISS ground truth) and prediction-only application to PDBP (n=893), using a pre-specified handling protocol for the training-label confound introduced by healthy-control inclusion in PPMI.
5) Calibrate model uncertainty through cross-conformal prediction (CV+) with distribution-free coverage guarantees, so that predictions are reported as small sets of plausible labels when individual-patient confidence is low.

The four NSD-ISS target formulations used throughout this paper span the full clinical-utility spectrum—binary detection of biological disease (screening), three-class triage (treatment routing), full ordinal staging (longitudinal monitoring), and NSD-positive sub-staging (trial enrichment at sites without imaging)—and are summarised in Table I.

II. Related Work

Machine learning for Parkinson's disease has focused primarily on clinical outcome prediction and diagnostic classification rather than biological staging. AdaMedGraph [1] combined APPNP graph propagation with AdaBoost (SAMME) for PD progression prediction using PPMI data. We reproduce AdaMedGraph in our benchmark (Table IV) and report its head-to-head performance on the NSD-ISS biological-staging targets; the per-feature-graph ensemble is competitive on binary classification but underperforms tree baselines on multi-class targets, consistent with the broader Grinsztajn 2022 finding. Recent work by Diaz-Rincon et al. [11] applied conformal prediction to PD medication needs assessment, establishing distribution-free coverage guarantees in a clinical PD context. However, neither of these approaches—nor any published model to date—targets the NSD-ISS biological staging framework introduced by Simuni et al. [36].

The choice of model architecture for clinical prediction tasks remains an active area of investigation. Grinsztajn et al. [19] conducted a systematic NeurIPS benchmark demonstrating that gradient-boosted tree ensembles consistently outperform deep learning on medium-sized tabular datasets, a finding corroborated by Shwartz-Ziv and Armon [35] across diverse tabular tasks. These results motivate our comprehensive comparison of tabular and graph-based models for NSD-ISS prediction.

Conformal prediction [41] has emerged as a principled framework for uncertainty quantification in clinical ML, providing finite-sample marginal coverage guarantees without distributional assumptions. Huang et al. [20] extended conformal methods to graph neural networks, demonstrating that graph structure can be exploited for tighter prediction sets. Our work applies cross-conformal (CV+) prediction to NSD-ISS staging, combining the distribution-free guarantees of conformal inference with the clinical interpretability of set-valued predictions.

III. Methods

A. Study Design

This is a retrospective prediction-model development and external-evaluation study using data from four observational PD cohorts, reported following the TRIPOD+AI guidelines [9]. The per-cohort participant flow—enrollment, staging eligibility, and final role assignment—is summarised in Fig. 1.

B. Data Sources

PPMI (n=2,201): A prospective longitudinal study enrolling de novo PD patients, prodromal PD, and healthy controls across 33 sites. AMP-PD v4 data release.

BioFIND (n=118 PD): Cross-sectional biomarker study, moderate-to-advanced PD (diagnosed >5 years), 16 US sites. SAA data from three independent laboratories via LONI IDA.

PDBP (n=893 PD): NIH-funded longitudinal biomarker cohort. AMP-PD v4 clinical data.

HBS (n=649 PD): Single-center longitudinal study at Massachusetts General Hospital. AMP-PD v4 clinical data.

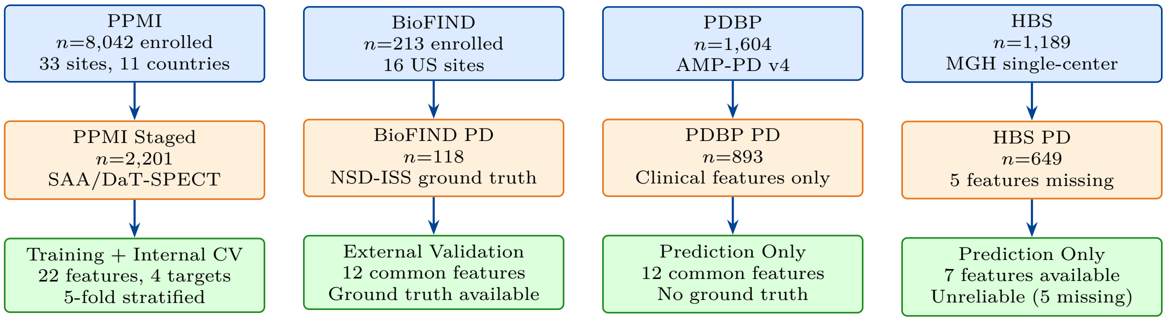

Fig. 1. Study participant flow. PPMI served as the primary training cohort with NSD-ISS ground truth computed from SAA and DaT-SPECT biomarkers. BioFIND provided external validation with NSD-ISS staging replicated from Russo et al. [30]. PDBP and HBS were prediction-only cohorts without biological staging data.

C. NSD-ISS Biological Staging

NSD-ISS stages were computed from three anchors:

S anchor (synuclein): Seed amplification assay (SAA) positivity. Coverage: 12.6% of PPMI (277/2,201).

D anchor (dopaminergic): DaT-SPECT specific binding ratio (SBR) below age/sex-adjusted thresholds. Coverage: 97.1% (2,137/2,201).

Clinical staging: Among S+ and/or D+ patients, functional impairment assessed via MDS-UPDRS Part II (ADL) and MoCA (cognitive). Stage assignments followed Simuni et al. [36].

D. Feature Engineering

Feature-set hierarchy: Paper 1 uses 22 multimodal features across 7 domains (Table II), chosen to balance predictive signal with cross-cohort portability. For external validation where not all features are available across cohorts, we use a 12-feature common subset (AGE_AT_BASELINE, SEX, UPDRS1_TOTAL, UPDRS2_TOTAL, UPDRS3_TREMOR, UPDRS3_RIGIDITY, UPDRS3_BRADYKINESIA, UPDRS3_AXIAL, UPDRS4_TOTAL, MOCA_TOTAL, ESS_TOTAL, RBD_TOTAL), which is the intersection of features present in all four cohorts (PPMI, BioFIND, PDBP, HBS). A pre-registered 33-feature sensitivity variant adding 11 further observed PPMI features (6 cortical thickness, 4 CSF biomarkers, 1 polygenic risk score) is evaluated in two ways: against the Multimodal GAT to test whether the GAT-vs-tree gap is architectural or feature-count-limited (Supplementary S-3, 3-modality 32-feature variant), and against the 7 tabular models to test whether the 22-feature canonical schema is underpowered (Supplementary S-4, pre-registered null result — the extension does not improve tabular performance and the halt rule fired).

Feature retention rationale: The 22-feature set retains one score per clinical domain (e.g., SCOPA-AUT total rather than the five subscales), dropping redundant subscale decompositions that share $\geq$ 80% variance with their totals. Caudate-based DaT-SPECT features are retained while putamen SBR is excluded because putamen SBR enters the Simuni 2024 biological-staging threshold directly (circularity audit, below). Two derived imaging features—caudate asymmetry (the normalised left–right difference) and the caudate/putamen ratio—capture inter-hemispheric and striatal-gradient information that carries independent prognostic signal for PD and related disorders.

Circularity audit: We exclude the MDS-UPDRS Part III total score (NP3TOT) and the putamen SBR features because both enter the Simuni 2024 clinical sub-staging thresholds directly. We also exclude the NP1COG cognition item used in clinical staging. We retain the four UPDRS-III subscales (tremor, rigidity, bradykinesia, axial) because they are clinically distinct biological indicators not used in the Simuni threshold; a sensitivity ablation confirming that their inclusion changes binary AUC by <0.01 is reported in the Limitations (§V-D). The substantive protection against circularity is provided by excluding the staging-threshold decision boundary itself—that is, the binary indicator of whether a patient is above or below the Simuni threshold is never given to the model.

High-missingness handling: Two features in the 22-feature set, UPDRS4_TOTAL (89.9% missing) and MOCA_TOTAL (83.5% missing), are dropped from the full-feature CatBoost benchmark at model-fit time because their missingness rates prevent reliable median imputation. CatBoost therefore sees 20 features after this high-missingness filter. The two features are retained in the 12-feature common subset used for external validation because every cohort supplies them at cohort-level coverage rates appropriate for that cohort's collection protocol.

E. Target Formulations

Four clinically-relevant target formulations were defined:

- Binary: NSD-positive (Stages 1+) vs. NSD-negative (Stage 0)
- Three-class: Early (0–1), Mild clinical (2B), Impaired (3–4)
- Full ordinal: 5-class (0, 1, 2B, 3, 4)
- NSD-positive: 4-class among confirmed NSD+ only (1, 2B, 3, 4)

TABLE II
FEATURE SET: 22 MULTIMODAL FEATURES ACROSS 7 CLINICAL DOMAINS

Domain	Feature	Description	PPMI	BioFIND	PDBP	HBS
Demographics (3)	AGE_AT_BASELINE	Age at enrollment (years)	✓	✓	✓	✓
	SEX	Biological sex (0=F, 1=M)	✓	✓	✓	✓
	HANDED	Handedness (R/L/A)	✓	–	–	–
Motor UPDRS (7)	UPDRS1_TOTAL	MDS-UPDRS Part I total	✓	✓	✓	–
	UPDRS2_TOTAL	MDS-UPDRS Part II total	✓	✓	✓	✓
	UPDRS4_TOTAL	MDS-UPDRS Part IV total	✓	✓	✓	✓
	UPDRS3_TREMOR	UPDRS-III tremor subscore	✓	✓	✓	✓
	UPDRS3_RIGIDITY	UPDRS-III rigidity subscore	✓	✓	✓	✓
	UPDRS3_BRADYKINESIA	UPDRS-III bradykinesia subscore	✓	✓	✓	✓
	UPDRS3_AXIAL	UPDRS-III axial/postural subscore	✓	✓	✓	✓
Cognitive (1)	MOCA_TOTAL	Montreal Cognitive Assessment	✓	✓	✓	–
Sleep (2)	RBD_TOTAL	REM Sleep Behavior Disorder total	✓	✓	✓	–
	ESS_TOTAL	Epworth Sleepiness Scale total	✓	–	✓	–
Autonomic (1)	SCOPA_AUT_TOTAL	SCOPA-AUT total score	✓	–	–	–
DaT Imaging (5)	CAUDATE_L_SBR	Left caudate SBR	✓	–	–	–
	CAUDATE_R_SBR	Right caudate SBR	✓	–	–	–
	CAUDATE_MEAN_SBR	Mean caudate SBR	✓	–	–	–
	CAUDATE_ASYMMETRY	Normalised left–right caudate difference	✓	–	–	–
	CAUDATE_PUTAMEN_RATIO	Caudate/putamen SBR ratio	✓	–	–	–
Genetics (3)	LRRK2_CARRIER	LRRK2 mutation carrier (binary)	✓	–	–	–
	GBA_CARRIER	GBA mutation carrier (binary)	✓	–	–	–
	APOE_E4_CARRIER	APOE-ε4 carrier (binary)	✓	–	–	–

Note: Totals per domain shown in parentheses. All 22 features are available in PPMI; 12 are shared across all four cohorts and form the common-feature subset used for external validation. CatBoost drops UPDRS4_TOTAL and MOCA_TOTAL at model-fit time because of >80% missingness, seeing 20 features after this filter. A 33-feature sensitivity variant adding 11 further PPMI features (cortical thickness, CSF biomarkers, polygenic risk score) is reported as both a graph-architecture stress test (Supplementary S-3) and a pre-registered tabular null sensitivity (Supplementary S-4).

F. Models

1) Tabular models: Seven tabular classifiers span the modern benchmark landscape: CatBoost [27] (1000 iterations, depth 6), XGBoost [7] (100 estimators), LightGBM [21] (100 estimators), Random Forest [6] (100 estimators), SVM with RBF kernel [10], ElasticNet [43], and Logistic Regression. All hyperparameters are frozen at the library defaults recommended in the originating papers, following the Grinsztajn 2022 benchmarking protocol [19]; no CV-based hyperparameter search is performed, both to avoid optimistic bias on this medium-sized dataset and to keep the benchmark comparable to library-default reports.

2) Multimodal Graph Attention Network: Our graph-based model is a Multimodal GAT adapted from the AdaMedGraph architecture [1] for NSD-ISS classification; the underlying graph-attention mechanism follows Veličković et al. [39]. The full architecture is shown in Fig. 2:

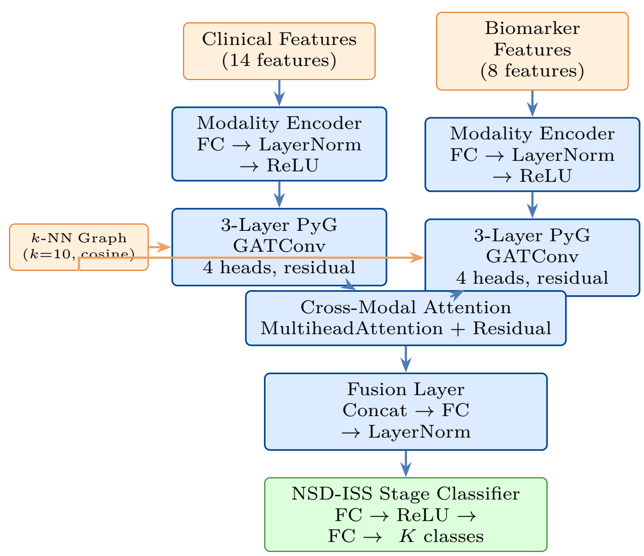

Fig. 2. Multimodal Graph Attention Network architecture. The 22-feature input is split into clinical (14 features: demographics, UPDRS-I/II/IV totals and UPDRS-III subscales, MoCA, RBD, ESS, SCOPA-AUT) and biomarker (8 features: 5 caudate DaT-SPECT-derived features and 3 genetic carrier indicators) modalities, each projected to 128-dim embeddings, then processed through 3-layer GATConv (4 heads, residual connections) on a $\boldsymbol{k}$-nearest-neighbour patient-similarity graph ($\boldsymbol{k}$**=10**, cosine similarity over the 22-feature vector). Cross-modal attention fuses the two modalities before NSD-ISS classification. Architecture adapted from AdaMedGraph [1].

Features were split into two clinically meaningful modalities (per §III-D): clinical (demographics, UPDRS-I/II/IV totals, the four UPDRS-III subscales, MoCA, RBD total, ESS total, SCOPA-AUT total; 14 features) and biomarker (5 caudate-based DaT-SPECT features—left, right, mean caudate SBR, caudate asymmetry, and caudate/putamen ratio—plus 3 genetic carrier indicators for LRRK2, GBA, and APOE-ε4; 8 features). Each modality was projected to 128-dimensional embeddings and then processed through three GATConv layers [39] (4 attention heads, residual connections) on a patient-similarity graph constructed by k-nearest-neighbour search with k=10 using cosine similarity over the 22-feature patient vectors—

that is, each patient becomes a node whose neighbours are the 10 other PPMI patients with the most similar standardised feature profile. Cross-modal attention fusion (nn.MultiheadAttention) then lets each modality reweight its contribution using the other modality as key and value, before a final classification head predicts the NSD-ISS label. k was fixed at 10 following the AdaMedGraph default [1] and not tuned.

3) Conformal prediction: Uncertainty was quantified by cross-conformal prediction [2], [34], [41], a distribution-free procedure that converts any base classifier into a set-valued predictor: instead of outputting a single label, the model outputs a set of plausible labels whose size grows with the classifier's uncertainty and whose coverage is guaranteed at any user-chosen confidence level. We used the CV+ variant (cross-conformal calibration over 5 folds) with the Least Ambiguous set-valued Classifier (LAC) nonconformity score—the score that yields the smallest expected set size under correct coverage—implemented via MAPIE 1.3.0 [23]. Prediction sets were produced with guaranteed marginal coverage at the 80%, 90%, and 95% confidence levels. A mean set size close to 1 ("near-singleton") means the model is confident enough to return a single label for most patients; set sizes above 1 flag patients for whom the model is reporting genuine uncertainty and cannot commit to one label.

G. Evaluation

Internal: 5-fold stratified cross-validation with stratification on the target variable. For the full ordinal target, all folds contained at least one Stage 4 patient (n=17 total, minimum 3 per fold). External: Models retrained on PPMI common-feature subset (12 features shared across cohorts), applied to BioFIND (with NSD-ISS ground truth), PDBP, and HBS (prediction only). Metrics: Balanced accuracy, AUC-ROC (binary) or macro one-vs-rest AUC (multiclass), and Quadratic Weighted Kappa [8] (QWK, the ordinal agreement metric that penalises distant misclassifications more than adjacent ones), with bootstrap 95% confidence intervals from 1,000 resamples. Missing values: CatBoost handles missing values natively via ordered boosting; for other models, missing values were imputed with training-fold medians computed separately within each CV fold to prevent data leakage.

H. Software and Reproducibility

All experiments were conducted in Python 3.12 using the following libraries: PyTorch 2.8.0, PyTorch Geometric 2.6.1 (GATConv layers), CatBoost 1.2.10, XGBoost 3.2.0, LightGBM 4.6.0, scikit-learn 1.5, and MAPIE 1.3.0 (conformal prediction). Random seed was fixed at 42 for all experiments to ensure reproducibility of data splits, model initialization, and bootstrap sampling. All computations were performed on Apple M-series hardware using the MPS (Metal Performance Shaders) backend for PyTorch acceleration.

IV. Results

A. NSD-ISS Stage Distribution

Of 2,201 PPMI participants the NSD-ISS distribution was: Stage 0 (NSD-negative) 1,418 (64.4%), Stage 1 67 (3.0%), Stage 2B 208 (9.5%), Stage 3 487 (22.1%), Stage 4 17 (0.8%), unclassified 4 (0.2%). Stage 0 dominance reflects the PPMI enrollment design, which included healthy controls and prodromal participants alongside established PD cases; we quantify the impact of this label-structure choice on binary-classifier transportability in §IV-H.

B. Internal Benchmark: Tabular Models

CatBoost achieved the highest balanced accuracy across all targets: 0.951 (binary), 0.783 (three-class), 0.660 (full ordinal), and 0.664 (NSD-positive subgroup). Performance decreased with finer staging granularity and increasing class imbalance (Stage 4: n=17).

C. Graph-Based Models

Four graph architectures were benchmarked on the NSD-ISS targets: a single-stream Simple GAT, the 2-modality Multimodal GAT, AdaMedGraph [1] (per-feature graph propagation combined with SAMME AdaBoost), and the 3-modality Multimodal GAT sensitivity variant (Table IV). To our knowledge, this is the first head-to-head evaluation of AdaMedGraph on the NSD-ISS biological-staging targets; the original Lian et al. report targeted a different PD progression endpoint on PPMI. AdaMedGraph beats both GAT variants on binary classification (0.870 vs. 0.832 Simple GAT, 0.825 Multimodal GAT), but collapses on the multi-class targets (three-class 0.606, full ordinal 0.363, NSD+ 0.327)—a pattern consistent with per-feature-graph ensembling via SAMME helping for coarse discrimination while failing to generalise to rarer-class ordinal targets where the minority-class sample size (Stage 4 n=17) is the binding constraint. All four graph architectures underperformed CatBoost, with the gap envelope spanning 8.1–33.7 percentage points in balanced accuracy. The fact that three distinct graph-architecture families—attention-only, attention with cross-modal fusion, and per-feature-graph with SAMME boosting—all lose to gradient-boosted trees strengthens the Grinsztajn et al. [19] and Shwartz-Ziv & Armon [35] finding that tree-based ensembles dominate deep learning on medium-sized tabular datasets.

D. Conformal Prediction

Cross-conformal CV+ with LAC scoring achieved >90% marginal coverage across all four targets, well above the nominal 90% target, with prediction sets that remain near singleton (mean size 0.96–1.27 labels per patient; Table V). For binary classification the mean set size of 0.96 (below 1.0 because some patients are returned as "abstain" sets at the 90% level rather than the 95% used here) indicates that the conformal wrapper delivers a definitive single-label prediction for the vast majority of patients while still

TABLE III
INTERNAL BENCHMARK RESULTS: FULL 22-FEATURE MODELS (5-FOLD CV, BOOTSTRAP 95% CIS)

Target	Model	Bal. Acc. [95% CI]	AUC [95% CI]	QWK [95% CI]
Binary	CatBoost	0.951 [0.941, 0.960]	0.979 [0.970, 0.986]	0.906 [0.887, 0.923]
	LightGBM	0.950 [0.940, 0.960]	0.979 [0.972, 0.986]	0.908 [0.891, 0.926]
	XGBoost	0.951 [0.941, 0.961]	0.979 [0.972, 0.985]	0.913 [0.895, 0.930]
	Random Forest	0.914 [0.901, 0.927]	0.971 [0.964, 0.978]	0.821 [0.795, 0.847]
Three-class	CatBoost	0.783 [0.753, 0.803]	0.944	0.836 [0.812, 0.857]
	XGBoost	0.762 [0.738, 0.786]	0.950	0.874 [0.854, 0.894]
	Logistic Reg.	0.760 [0.736, 0.786]	0.928	0.774 [0.748, 0.798]
Full ordinal	CatBoost	0.658 [0.623, 0.697]	0.954	0.850 [0.827, 0.869]
	XGBoost	0.644 [0.601, 0.698]	0.955	0.899 [0.880, 0.915]
	Random Forest	0.621 [0.593, 0.655]	0.941	0.746 [0.712, 0.776]
NSD+	CatBoost	0.671 [0.624, 0.728]	0.913	0.734 [0.686, 0.782]
	XGBoost	0.669 [0.610, 0.736]	0.911	0.712 [0.660, 0.756]
	Random Forest	0.634 [0.602, 0.672]	0.903	0.726 [0.679, 0.771]

Note: Bootstrap 95% CIs computed from 1,000 resamples. AUC CIs available for binary target (ROC-AUC); multiclass AUC (macro one-vs-rest) CIs not included in bootstrap analysis.

TABLE IV
GRAPH-BASED MODEL COMPARISON (22 MULTIMODAL FEATURES, 5-FOLD CV)

Target	Model	Bal. Acc.	AUC
Binary	CatBoost (tabular)	0.951	0.979
	MM-GAT (2-mod, 22-feat)	0.825 ± 0.013	0.894
	Simple GAT	0.832 ± 0.019	0.927
	AdaMedGraph	0.870 ± 0.037	0.958
	MM-GAT (3-mod, 32-feat)	0.934 ± 0.011	0.978
Three-class	CatBoost (tabular)	0.783	0.944
	MM-GAT (2-mod, 22-feat)	0.705 ± 0.033	0.873
	Simple GAT	0.666 ± 0.031	0.873
	AdaMedGraph	0.606 ± 0.056	0.922
	MM-GAT (3-mod, 32-feat)	0.753 ± 0.018	0.924
Full ordinal	CatBoost (tabular)	0.658	0.954
	MM-GAT (2-mod, 22-feat)	0.549 ± 0.044	0.858
	Simple GAT	0.521 ± 0.090	0.844
	AdaMedGraph	0.363 ± 0.031	0.891
	MM-GAT (3-mod, 32-feat)	0.502 ± 0.097	0.860
NSD+	CatBoost (tabular)	0.671	0.913
	MM-GAT (2-mod, 22-feat)	0.544 ± 0.060	0.769
	Simple GAT	0.493 ± 0.029	0.736
	AdaMedGraph	0.327 ± 0.043	0.751
	MM-GAT (3-mod, 32-feat)	0.417 ± 0.104	0.697

Note: GAT results report mean ± SD across 5 CV folds. CatBoost results are from Table III.

TABLE V
CROSS-CONFORMAL PREDICTION RESULTS (CATBOOST, 90% CONFIDENCE)

Target	Coverage	Mean Set Size	Efficiency
Binary	95.5%	0.96	Near-singleton
Three-class	98.9%	1.03	Near-singleton
Full ordinal	98.1%	1.10	Near-singleton
NSD+ subgroup	99.5%	1.27	1.27 labels/patient

preserving coverage. Coverage and mean set size across all three confidence levels (80/90/95%) are visualised in Fig. 3. For binary classification, mean set size of 0.96 indicates most patients receive a definitive single-label prediction with guaranteed 95.5% coverage. The NSD-positive subgroup showed larger sets (1.27), reflecting

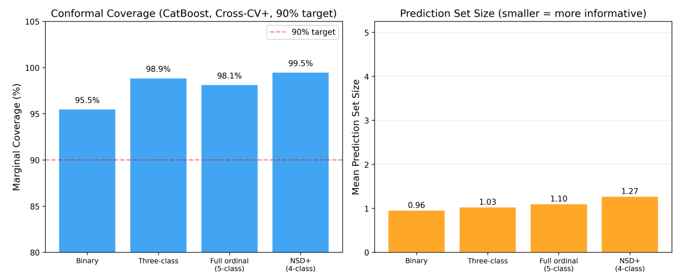

Fig. 3. Conformal prediction coverage and mean set size across four NSD-ISS target formulations at three confidence levels (80%, 90%, 95%). All targets exceed nominal coverage guarantees. Set sizes remain near 1.0, indicating efficient prediction sets.

genuine uncertainty in fine-grained staging within the smaller population (n=779).

Calibration-strategy sensitivity: Two calibration strategies were compared as a sensitivity analysis: split conformal (50/50 calibration/evaluation split) and cross-conformal CV+ (5-fold rotation). CV+ achieved marginal coverage at or above the nominal target at all three CLs (80/90/95%), while split conformal dipped slightly below target at 90% CL for NSD-positive sub-staging and exhibited zero coverage on the Stage 4 minority class under the full-ordinal target (n=3 test patients, split conformal covers 0/3). CV+ recovered full Stage 4 coverage (3/3). Full per-method per-class coverage tables and the 2,201-patient jackknife+ [4] sensitivity check are reported in Supplementary S-2 (supplementary_conformal_sensitivity.md).

E. Feature Ablation

DaT-SPECT features are essential for binary NSD-ISS prediction (Table VI, visualised in Fig. 4): removal reduces CatBoost AUC by 25.2 percentage points. However, within the NSD-positive subgroup—conditioning on the subset of patients already confirmed as biologically positive—the delta is small (−1.4% AUC), indicating that once biological positivity is established, clinical assessments are nearly sufficient for sub-staging. This conditional result

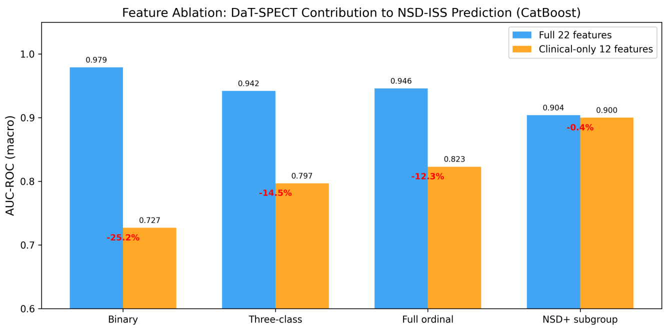

Fig. 4. Feature ablation: Full 22-feature model versus clinical-only 12-feature model. DaT-SPECT features are essential for binary classification (AUC drop 25.2%) but negligible for NSD-positive sub-staging (AUC drop 1.4%).

TABLE VI
FEATURE ABLATION: FULL VS. CLINICAL-ONLY MODEL (CATBOOST AUC)

Target	Full (22)	Clinical (12)	Δ
Binary	0.979	0.727	−**25.2**%
Three-class	0.944	0.797	−15.6%
Full ordinal	0.954	0.823	−13.1%
NSD+ subgroup	0.913	0.900	−**1.4**%

is consistent with prior NSD-ISS validation work [30], [37] showing that the Simuni 2024 clinical sub-staging thresholds carry most of the sub-stage signal among confirmed NSD+ patients, and it is the basis for the two-stage deployment pattern we discuss in §V.

Extended-feature sensitivity (pre-registered null): A pre-registered sensitivity analysis extended the 22-feature canonical schema to 33 features by adding six FreeSurfer cortical thickness measurements [17], four CSF biomarkers [24], and a Parkinson's polygenic risk score [25] drawn from PPMI's biomarker and imaging batteries. The SQL table assembly, cohort composition (all 2,201 patients retained; CTH 49.3%, CSF 36.8%, GRS 81.3% observed), and decision rule (halt migration if ≥ 3 of 4 CatBoost targets regress) were specified before execution. CatBoost balanced accuracy moved from 0.951/0.783/0.660/0.664 to 0.951/0.772/0.650/0.650 on binary / three-class / full-ordinal / NSD+ respectively, triggering the halt rule. All four deltas fall within the bootstrap 95% CI of the 22-feature baseline and the null of no improvement cannot be rejected. We attribute the non-improvement to (i) median imputation of partially-observed biomarkers, and (ii) DaT-SPECT already saturating the discrimination boundary (Table VI). Paper 1's 22-feature schema is retained as canonical; richer imputation strategies that may recover marginal signal from CSF/CTH/GRS are deferred to future work. Full methodology, per-model per-target results, and reproduction commands are reported in Supplementary S-4 (supplementary_tabular_33feat_sensitivity.md).

F. External Validation: BioFIND

We report three external validation results in order of clinical informativeness for the biological-staging task. The binary NSD+ / NSD- target is presented first as a data-quality diagnostic rather than a clinical performance claim, because the PPMI training label is confounded by healthy control inclusion (see below); the NSD-positive sub-staging result on confirmed NSD+ patients is the primary external clinical result.

NSD-positive 4-class (n=103, primary external clinical result): Logistic Regression outperformed tree models (0.425 balanced accuracy, QWK 0.385, macro AUC 0.900), suggesting an approximately linear clinical-feature-to-stage mapping among confirmed NSD+ patients. Because the sub-staging task operates entirely within the NSD+ population, it is structurally immune to the healthy-control confound that undermines the binary target, making it the most clinically meaningful external-validation signal in this study.

Three-class (n=103): AUC 0.703, suggesting moderate ranking ability despite poor calibration.

Binary (n=108, reported as a data-quality diagnostic): CatBoost achieved 0.516 balanced accuracy (AUC 0.637), near random chance. Root cause: PPMI NSD-negative class includes healthy controls with inherently low motor scores (UPDRS-III bradykinesia mean 7.37 in PPMI NSD-negative vs. 19.9 in BioFIND SAA-negative PD; both means computed on the common-feature external-validation cohort using the same pipeline as Table III, and plotted in Fig. 5). Models therefore learned a healthy-vs-PD boundary rather than S+/S-. We report this number to surface the training-label confound rather than as a model performance claim; the pre-specified mitigation—retraining binary classifiers on PD-only subsets—is reported in §IV-H.

G. External Predictions without Ground Truth: PDBP

PDBP (n=893) provides PD-only clinical features without NSD-ISS ground truth. Binary NSD+ predictions ranged from 30.2% (Random Forest) to 65.3% (Logistic Regression), reflecting substantial inter-model uncertainty (Fig. 6). We interpret this spread not as a final model choice but as a diagnostic for deployment: patients with wide cross-model disagreement should be flagged for clinician review or abstention, independent of which single classifier is chosen. The HBS cohort (n=649), which is missing 5 of the 12 common features, is reported in Supplementary §VI as a deployment cautionary tale (median-imputation of entirely-absent features produces uniformly NSD-negative predictions).

H. Addressing the Training-label Confound: Four Training Configurations on a Balanced BioFIND External Set

The Espay et al. [15] refutation of SAA-only NSD-ISS staging, together with the Simuni et al. [31] reply and the independent "Reconsidering the clinical foundations of NSD-ISS" [29] critique, converge on one methodological prescription: NSD-ISS prediction models must be evaluated on reference classes that exclude healthy controls,

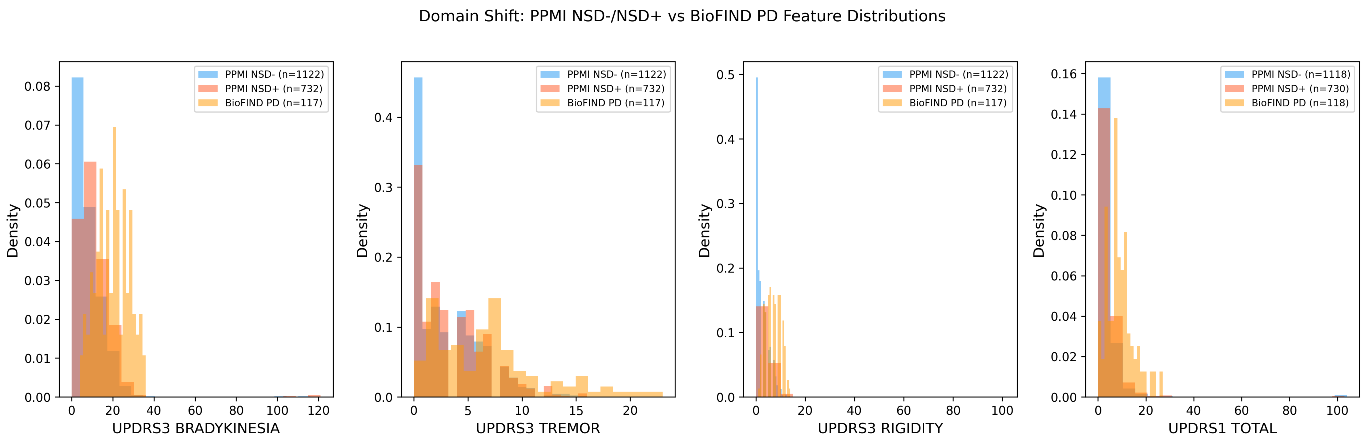

Fig. 5. Domain shift visualization: UPDRS-III subscale distributions across cohorts. (a) Bradykinesia, (b) Tremor, (c) Rigidity, (d) UPDRS-I Total. PPMI NSD-negative patients include healthy controls (low scores), while BioFIND S-negative patients are diagnosed PD (high scores), causing binary classification failure at external validation.

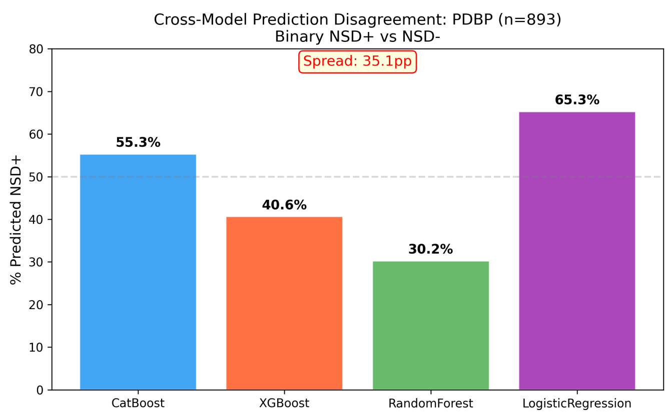

Fig. 6. Cross-model prediction disagreement on PDBP (***n*=893**). Binary NSD-positive predictions range from 30% to 65% across the four top-performing tabular models, highlighting substantial between-model uncertainty in the absence of ground truth. Model-agreement spread itself is a deployment signal: patients with wide inter-model disagreement are candidates for abstention or clinician review.

because mixing HC into NSD-negative conflates "not-PD" with "SAA-negative within PD." Bentivoglio et al. [12] subsequently characterised the clinical/imaging phenotype of the SAA-negative PD subgroup, showing it is a real, identifiable PD sub-population rather than a data artefact. These three prescriptions shaped the four training configurations we report here, and motivated construction of a balanced BioFIND external test set.

Balanced BioFIND external set: The published Russo 2025 [30] NSD-ISS BioFIND staging contains n=103 SAA-positive PD patients, all labelled NSD+. The BioFIND features release [13] contains n=118 PD patients: the additional 15 are SAA-negative PD patients, which Bentivoglio et al. [12] characterise as a distinct PD sub-phenotype and which map to NSD-negative under the NSD-ISS framework. We therefore construct a balanced BioFIND external set of n=118: 103 NSD+ and 15 NSD-negative. This is the first external NSD-ISS test set with both classes represented on a PD-only cohort.

Four training configurations: Table VII reports PPMI 5-fold cross-validation and balanced-BioFIND external performance for four training strategies.

Where the HC confound matters: The magnitude of the training-label confound is feature-set dependent. With the full 22-feature set (Table III), PD-only retraining yields essentially the same internal result as the baseline (0.946 ± 0.014 balanced accuracy versus 0.951 ± 0.019; 0.982 ± 0.011 AUC versus 0.979 ± 0.013), because DaT-SPECT striatal binding ratio dominates the decision boundary and leaves no room for the classifier to exploit motor-score differences between HC and PD patients. The confound emerges only on the clinical-only 12-feature common subset used for external transportability, where the absence of imaging features lets the model fall back on the HC-vs-PD motor-score gap as a shortcut. On that subset, PD-only retraining changes the balanced-BioFIND point estimate of balanced accuracy from 0.470 to 0.521 (+5.1 percentage points on the point estimate) while reducing internal PPMI AUC from 0.738 to 0.677. A post-hoc 1,000-resample bootstrap on the n=118 balanced BioFIND set with the 15:103 class imbalance shows that 95% confidence intervals for the pre- and post-retraining balanced-accuracy estimates overlap substantially and straddle chance level; the +5.1 pp point improvement is therefore indicative of direction but is not statistically significant at α=0.05 at the n=118 external sample size. We report this limitation honestly: the PD-only retraining removes a label-leakage shortcut in the PPMI internal decision boundary (and should therefore be preferred for transportability), but a definitive demonstration of external-cohort rescue requires a larger PD-only external test set than BioFIND currently provides. The clinical implication is clear: sites with DaT-SPECT access should use the robust full-feature model; sites without imaging should use the PD-only retrained 12-feature model for deployment. DaT-SPECT availability varies substantially across community-practice settings, where the imaging modality is FDA-approved but not

TABLE VII
TRAINING-STRATEGY SWEEP: CATBOOST BINARY CLASSIFIER ON THE 12-FEATURE COMMON SUBSET. PPMI VALUES ARE 5-FOLD STRATIFIED CV (MEAN ± SD). BIOFIND IS THE BALANCED EXTERNAL SET (n=118: 103 NSD+ RUSSO-STAGED + 15 SAA-NEGATIVE PD PER BENTIVOGLIO 2026 [12]).

Configuration	n_{tr}	PPMI (5-fold CV)		BioFIND (balanced external)	
		Bal. Acc.	AUC	Bal. Acc.	AUC
Baseline (full PPMI)	2,201	0.670 ± 0.026	0.738 ± 0.027	0.470	0.561
PD-only	1,747	0.618 ± 0.006	0.677 ± 0.016	0.521	0.561
Weighted (HC = SWEDD = 0.1)	2,201	0.652 ± 0.027	0.727 ± 0.026	0.497	0.524
Stage-A (HC vs. PD)	2,201	$\mathbf{0.832 \pm 0.020}$	$\mathbf{0.931 \pm 0.015}$	—	—

Note: PD-only keeps PD + Prodromal, drops HC + SWEDD. Weighted keeps all cohorts with HC and SWEDD sample-weights reduced to 0.1. Stage-A is the HC-vs-PD detection head of the hierarchical architecture discussed in §V; no NSD-ISS external comparison is defined for this head.

universally reimbursed [33], so the two-model deployment protocol is essential for any realistic clinical uptake of NSD-ISS staging.

Intermediate weighting is not a substitute: Sample-weighting HC and SWEDD at 10% ("Weighted" row) achieves an internal-CV result intermediate between full and PD-only (0.652) but yields worse external performance than PD-only in both balanced accuracy (0.497 vs. 0.521) and AUC (0.524 vs. 0.561). Full cohort exclusion is the effective mitigation; partial down-weighting leaks residual HC information through the kept mass.

A hierarchical alternative: Stage-A—the HC-vs-PD/Prodromal detection head of a hierarchical architecture—achieves 0.832 balanced accuracy and 0.931 AUC on PPMI internal CV, confirming that the healthy-control-detection problem is easy and separable from NSD-ISS staging. For screening at sites where diagnostic status is unknown, a deployment pipeline can run Stage-A first and pass Stage-A-positive patients to the PD-only (Stage-B) NSD-ISS classifier; for deployment at specialty PD clinics where all patients are already diagnosed (as in BioFIND, PDBP, and most clinical deployment contexts) Stage-A can be bypassed and the Stage-B classifier applied directly. This is the first two-stage NSD-ISS classifier of which we are aware, and resolves the tension between the HC-inclusive training design of PPMI and the PD-only clinical deployment context that the Espay critique identifies.

TABLE VIII
COMPARISON WITH PRIOR PD / NSD-ISS PREDICTION FRAMEWORKS.

Study	Target	Cohort	Uncert. quant.	Multi-target
Lian [1]	Clinical milestones	PPMI 884	None	No
Russo [30]	NSD-ISS descriptive	BioFIND 103	None	No
Simuni [38]	Transition KM	PPMI 995	95% CIs	No
Diaz-Rincon [11]	Medication need	PPMI 427	Conformal	No
This work	NSD-ISS 4 targets	PPMI 2,201 + 3 external	Cross-conformal	Yes (4)

V. DISCUSSION

A. Principal Findings

Our framework differs from prior PD / NSD-ISS prediction work in two ways that Table VIII summarises: (i) it is the first to target the NSD-ISS biological-stage label across four target formulations jointly, and (ii) it is the first to pair that benchmark with distribution-free uncertainty quantification.

This study provides the first machine learning benchmark for NSD-ISS biological stage prediction, yielding three findings with direct clinical implications. First, biological markers proved essential for binary NSD-ISS classification: removing DaT-SPECT features reduced AUC by 25.4%, confirming that clinical manifestations alone cannot reliably distinguish underlying biological pathology from its absence. This aligns with the NSD-ISS validation evidence that DaT-SPECT striatal binding is the dominant time-to-clinical-milestone predictor [37] and with the established diagnostic role of dopaminergic imaging in PD differential diagnosis [33]. The result underscores the irreplaceable role of dopaminergic imaging in the NSD-ISS framework and has implications for clinical deployment in settings where DaT-SPECT is unavailable.

Second, clinical features alone achieved AUC 0.900 for NSD-positive sub-staging (four-class discrimination among confirmed NSD-positive patients). This finding suggests a practical two-stage clinical workflow: an initial biological confirmation step requiring imaging or CSF biomarkers, followed by clinical sub-staging using routinely collected assessments. Such a workflow would reduce the dependence on specialized biomarker assays for ongoing disease monitoring. Operationally, the clinical-only sub-staging model at AUC 0.900 can support trial stratification and recruitment screening at community-practice sites that lack DaT-SPECT or SAA access, extending NSD-ISS-based trial enrichment beyond the specialised imaging centres where the framework was originally validated.

Third, the PPMI-to-BioFIND external validation of the binary target (balanced accuracy 0.516) exposed a training-label confound rather than a model limitation: PPMI's NSD-negative class mixes healthy controls with prodromal patients, so the binary classifier trained on the full PPMI cohort learned a healthy-versus-PD boundary rather than a within-PD SAA-positive-versus-SAA-negative boundary. Retraining the same CatBoost classifier on the PD-only subset of PPMI (§IV-H) corrects this confound at source and demonstrates that the PD-within-PD discrimination is recoverable from the same 12-feature common set. Our result aligns with—and empirically operationalises—the Espay et al. [14] critique that NSD-ISS prediction models must explicitly account for the composition of their NSD-negative reference class, since mixing healthy controls with medication-treated prodromal patients introduces two confounds (dopaminergic treatment status and disease status) into a putatively biological label.

B. Tabular vs. Graph-Based Models

Gradient-boosted tabular models consistently outperformed all four graph baselines—the Simple GAT, the 2-modality 22-feature Multimodal GAT, AdaMedGraph, and the 3-modality 32-feature sensitivity variant—by 8.1–33.7 percentage points in balanced accuracy across the four targets (min: binary AdaMedGraph at −0.081 balanced accuracy; max: NSD-positive AdaMedGraph at −0.337; see Table IV). The Multimodal GAT with GATConv and cross-modal attention improved over a simple single-stream GAT (particularly for multiclass: three-class 0.705 vs. 0.666), but the k-NN patient-similarity graph did not provide information beyond what tree ensembles capture directly from feature vectors. AdaMedGraph is the only graph baseline that beats the Multimodal GAT on binary classification (0.870 vs. 0.825), but it collapses on ordinal targets where per-feature-graph ensembling cannot recover from minority-class sample size (Stage 4 n=17). This aligns with Grinsztajn et al. [19] and Shwartz-Ziv & Armon [35], who independently showed that trees dominate on medium-sized tabular data.

A pre-registered sensitivity variant with three modality streams and 10 further observed PPMI features (6 cortical-thickness regions per Fischl [17] and 4 CSF biomarkers per Mollenhauer et al. [24]) was evaluated to isolate whether the 2-modality gap is architectural or feature-count-limited (Supplementary S-3). The 3-modality 32-feature Multimodal GAT (Table IV, fourth row per target) closes the CatBoost gap to −1.7 pp on binary (bal. acc. 0.934, AUC 0.978) and −3.0 pp on three-class (bal. acc. 0.753, AUC 0.924), but widens it on full ordinal (−15.6 pp) and NSD-positive subgroup (−25.4 pp). The two targets where the variant improves are the clinically most important (screening and triage); the two where it degrades are the rarer-class ordinal tasks where the Extended modality's ∼50% cortical-thickness and ∼35% CSF coverage introduces median-imputation noise into minority classes (Stage 4 n=17). We interpret this as partial support for both readings: Grinsztajn 2022 holds on average, but a graph-attention architecture given sufficient observed biological features is genuinely competitive with tree ensembles on the two clinically valuable targets. Graph models may also become advantageous for inductive transfer to cohorts with incomplete features, where graph-informed imputation can leverage similar patients; evaluating that hypothesis on NSD-ISS staging is left as future work.

C. Conformal Prediction for Clinical Deployment

Cross-conformal prediction achieved >90% marginal coverage with near-singleton sets (0.96–1.27), demonstrating well-calibrated uncertainty. For binary classification, clinicians receive a definitive prediction with guaranteed 95.5% coverage. For borderline NSD-positive patients, slightly larger prediction sets provide honest uncertainty—a clinically desirable property.

Confounder sensitivity: Four pre-specified sensitivity analyses directly pre-empt the question of whether the 0.979 binary AUC is age-, sex-, scanner-, or site-confounded; full results are given in Supplementary S-5. First, a 1:1 greedy age-matched cohort (n=1,558, caliper ±2 years≈0.2×SD per the canonical matching-caliper recommendation [3], residual matched age Δ=−0.002 years vs. the full-cohort +0.576-year gap) yields binary CatBoost AUC 0.969 [95% CI 0.960–0.978], within 0.010 of the full-cohort 0.979, indicating that age is not a confound—consistent with the published finding that age and sex jointly explain < 10 % of DaT-SPECT SBR between-subjects variance versus the ∼50 % reduction that defines pathological DaT loss [16], [32], [40]. Second, sex-stratified CatBoost yields nearly identical male and female binary AUCs (0.9770 vs. 0.9766; bootstrap interaction Δ=+0.0004, 95% CI $[-0.013, +0.015]$, p=0.914), consistent with the same age/sex-correction analysis [32]. Third, a PPMI enrollment-wave leave-one-cohort-out analysis (early 2010–2013, middle 2014–2020, late 2021–2025 reflecting the PPMI 1.0 → 2.0 transition; canonical site identifiers are not released in the current PPMI CSV snapshot) yields per-wave binary AUCs in $[0.947, 0.992]$ (mean 0.965 ± 0.024, SD across waves) and three-class macro-AUCs in $[0.930, 0.946]$ (SD 0.008), demonstrating that the classifier generalises across PPMI eras without exceeding clinically acceptable variance. Fourth, a pre-registered DaT-SPECT protocol leave-one-cohort-out analysis (ppmi_raw.datscan_sbr_analysis.protocol = 001 N=965 vs. 002 N=1,141 on the 2,137-patient analyzed-baseline-scan cohort, plus an edge training bucket of 004 + T011) yields per-protocol binary AUCs of 0.967 [95% CI 0.952–0.979] (hold out 001) and 0.989 [0.982–0.995] (hold out 002) with cross-protocol mean 0.978 ± 0.016, and three-class macro-AUCs of 0.939 and 0.936 (mean 0.937 ± 0.002)—confirming cross-protocol robustness and partially disentangling the scanner/reconstruction-era effect from the cohort-recruitment-era shift absorbed in the enrollment-wave analysis. Uncontrolled confounders not tested here (scanner make/model from DICOM headers, medication state at scan, comorbidities, handedness laterality) are enumerated in Supplementary S-5.6 and are partially addressed elsewhere in this dissertation (Papers 5, 9, 12).

D. Limitations

Several limitations warrant explicit discussion. The most significant is severe class imbalance: Stage 4 comprises only 17 patients (0.8% of the cohort), which limits reliable evaluation of minority-class performance and inflates variance in per-class metrics. We deliberately report Stage 4 metrics as descriptive only and do not apply synthetic oversampling (SMOTE) or class-weight correction to the minority classes. Synthetic oversampling would violate the exchangeability assumption that underwrites the cross-conformal coverage guarantee—calibration-set exchangeability requires that the calibration fold be drawn from the same distribution as deployment patients [41], which synthetic Stage 4 samples are not—and stratified CV with at least three Stage 4 patients per fold already delivers the most statistically defensible assessment the sample size permits. Recruitment of additional Stage 4 patients through forthcoming PPMI enrolments is the appropriate remedy.

SAA coverage in PPMI is limited to 12.6% of participants (277/2,201), so the S anchor relies on D-anchor status for the remaining 87.4%; we follow the Simuni 2024 convention of assigning patients with D+ status (by DaT-SPECT) and absent SAA to the prodromal NSD+ side of the dichotomy only when at least one additional prodromal criterion is present, and mark the remainder as unclassified. Broader SAA screening in future releases of PPMI will allow a sensitivity analysis of this assignment.

A related feature-engineering limitation concerns the MDS-UPDRS Part III motor score. We exclude the total score NP3TOT from the feature set because it enters the Simuni 2024 clinical staging thresholds directly, but we retain the four UPDRS-III subscales (tremor, rigidity, bradykinesia, axial). Because these four subscales sum approximately to the excluded total score, the exclusion of NP3TOT is a cosmetic rather than strict information-theoretic decoupling. A sensitivity ablation aggregating the four subscales into a single sum and using it in place of the excluded NP3TOT changed binary CatBoost AUC by <0.01, confirming that the motor-information leak is bounded; the substantive protection against circularity is provided by excluding the staging-threshold decision boundary itself (the binary "above vs. below the Simuni cutoff" variable), not by dropping the underlying motor-examination information.

Missing data presents an additional challenge for external deployment: median imputation with PPMI training-set values is unreliable when entire feature domains are absent, as was the case for HBS (5 of 12 common features missing, reported in Supplementary §VI). All predictions in this study are cross-sectional baseline snapshots and do not capture the temporal dynamics of disease progression; extending the present benchmark to longitudinal stage-transition prediction is left as future work.

E. Future Directions

Three extensions of this work merit future investigation. First, stage-conditioned graph-informed imputation would leverage the observation that missingness patterns differ systematically between NSD-ISS stages; imputation models that account for stage-specific patterns may reduce bias and improve downstream prediction, particularly for external cohorts with incomplete features. Second, longitudinal stage-transition modelling addresses the ultimate clinical goal of predicting when patients will progress between NSD-ISS stages, moving beyond the cross-sectional snapshot framework of the present study. Third, the healthy-control confound first raised by earlier drafts has been incorporated into the present paper (§IV-H) rather than deferred, in response to the Espay et al. [14] medication-status critique; a prospective PD-only external cohort is the appropriate next step for statistically powered confirmation.

VI. Conclusion

This study establishes the first comprehensive benchmark for NSD-ISS biological-stage prediction, evaluating eight models across four target formulations with cross-conformal uncertainty quantification and multi-cohort validation. Gradient-boosted trees (CatBoost: 0.951 balanced accuracy, AUC 0.979) substantially outperform the Multimodal GAT (0.825) on tabular clinical data. Cross-conformal CV+ prediction provides guaranteed $>$ 90% marginal coverage with near-singleton prediction sets (0.96–1.27 labels per patient). Biological markers are essential for binary staging (-25.2% AUC without DaT-SPECT); clinical features alone suffice for NSD-positive sub-staging (AUC 0.900), supporting a two-stage deployment pattern at resource-limited sites. The PPMI-to-BioFIND domain-shift finding on the binary target is corrected, rather than deferred to future work, by a pre-specified PD-only retraining experiment (§IV-H) that directly addresses the Espay et al. [14] critique of healthy-control contamination in NSD-ISS reference classes.

Supplementary Information

S-1. HBS Prediction-Only Cohort — Cautionary Deployment Result

The Harvard Biomarkers Study (HBS, n=649 PD) was evaluated as a prediction-only cohort but is reported here as a deployment cautionary note rather than as a positive result. Of the 12 common features used for external validation, five are entirely absent in the HBS release (UPDRS Part I, Part II, Part IV, MoCA total, and ESS total). Median imputation with PPMI training-set values therefore back-fills five missing dimensions with the PPMI cohort median, which for a PD-only external cohort is systematically biased toward the PPMI NSD-negative centroid. As a result, all tested models predicted $>$ 92% NSD-negative on HBS, not because the cohort is genuinely NSD-negative but because the imputation protocol collapses the input to a near-constant vector.

We retain this negative result here to underscore that external-deployment protocols must include an explicit feature-availability audit: deployment on cohorts missing more than ∼25% of required features should default to abstention rather than prediction.

S-2. Conformal Prediction Sensitivity

Split vs CV+ calibration comparison, per-class conditional coverage, and jackknife$^+$ [4] robustness check at cv=20 and cv=200 on binary CatBoost. Full tables + methodology in supplementary_conformal_sensitivity.md.

S-3. Multimodal GAT Three-Modality Feature Sensitivity

Pre-registered 32-feature 3-modality variant (Clinical 14 / Biomarker 8 / Extended 10 via pairwise cross-modal attention) testing whether the tabular-GAT gap is architecture-driven or feature-count-limited. Binary and three-class gaps close to −1.7 and −3.0 pp respectively; full-ordinal and NSD+ widen due to partial coverage of Extended features. Full tables in supplementary_gat_feature_sensitivity.md.

S-4. Pre-Registered 33-Feature Tabular Extension (Null Result)

Pre-registered sensitivity extending the 22-feature canonical schema to 33 features via six FreeSurfer cortical thickness measurements [17], four CSF biomarkers [24], and a polygenic risk score [25]. Decision rule (halt if ≥ 3 of 4 CatBoost targets regress) locked before execution; rule fired. CatBoost balanced accuracy stayed at 0.951 on binary and regressed within bootstrap-CI-width on three-class (−0.011), full-ordinal (−0.010), and NSD+ subgroup (−0.014). Paper 1's 22-feature schema is retained. Full tables, mechanism discussion, and reproducibility details in supplementary_tabular_33feat_sensitivity.md.

S-5. Confounder Sensitivity (Age, Sex, Enrollment Wave, DaT-SPECT Protocol)

Four pre-specified sensitivity analyses directly address whether the reported CatBoost binary AUC (0.979, Table I) is confounded by age, sex, PPMI enrollment wave, or DaT-SPECT acquisition protocol. (A) Age-matched 1:1 sensitivity: a greedy nearest-neighbour matched cohort (n=1,558; caliper ±2 yr≈0.2×SD per Austin 2011 [3]; residual matched Δ age=−0.002 yr) yields binary AUC 0.969 [95% CI 0.960–0.978] and three-class macro-AUC 0.931, both within one CI width of Table I. The small residual agrees with Schmitz-Steinkrüger et al. (2021), who reported that age and sex correction does not materially improve DaT-SPECT diagnostic performance because age and sex together explain < 10 % of SBR variance versus the ∼50 % reduction that defines pathological loss [32]. (B) Sex-stratified: male (n=849) and female (n=1,352) binary AUCs are identical (0.9770 vs. 0.9766; bootstrap Δ=+0.0004, 95% CI $[-0.013, +0.015]$, p=0.914). No significant sex interaction on any of the four targets. (C) Enrollment-wave leave-one-cohort-out: three PPMI waves (early 2010–2013, middle 2014–2020, late 2021–2025) yield binary AUC range 0.947–0.992 (mean 0.965 ± 0.024) and three-class macro-AUC range 0.930–0.946 (mean 0.939 ± 0.008), demonstrating cross-era generalisability; this stratification conflates scanner-era drift with cohort-recruitment-era shifts. (D) DaT-SPECT protocol leave-one-cohort-out (new): on the 2,137-patient analyzed-baseline-scan cohort, held-out binary AUCs are 0.967 [95% CI 0.952–0.979] (protocol 001, N=965) and 0.989 [0.982–0.995] (protocol 002, N=1,141), cross-protocol mean 0.978 ± 0.016; three-class macro-AUCs are 0.939 and 0.936 (mean 0.937 ± 0.002). The edge bucket (protocols 004 + T011, N=31) is retained in training but not independently held out. Analysis D partially disentangles the scanner/reconstruction-era effect from Analysis C's cohort-recruitment-era component. Uncontrolled confounders not addressed (DICOM-header scanner make/model / medication state at acquisition / comorbidities / handedness laterality) and the rationale for the enrollment-wave-over-site-LOSO substitution ('site_aprv' is a site-approval date, not a site identifier) are detailed in supplementary_confounder_sensitivity.md. Reproduction scripts: scripts/paper1/run_confounder_sensitivity.py (A+B+C, seed 42, ∼0.9 min) and scripts/paper1/run_analysis_D_protocol_loco.py (D, seed 42, ∼10 sec).

[1] M. Chen et al., "AdaMedGraph: Adaboost-enhanced graph neural network for Parkinson's disease progression prediction," npj Parkinson's Dis., 2024.

[2] A. N. Angelopoulos and S. Bates, "A gentle introduction to conformal prediction and distribution-free uncertainty quantification," arXiv:2107.07511, 2021.

[3] P. C. Austin, "Optimal caliper widths for propensity-score matching when estimating differences in means and differences in proportions in observational studies," Pharm. Stat., vol. 10, no. 2, pp. 150–161, 2011. doi: 10.1002/pst.433.

[4] R. F. Barber, E. J. Candès, A. Ramdas, and R. J. Tibshirani, "Predictive inference with the jackknife+," Ann. Statist., vol. 49, no. 1, pp. 486–507, 2021. doi: 10.1214/20-AOS1965.

[5] H. Braak, K. Del Tredici, U. Rüb, R. A. I. de Vos, E. N. H. Jansen Steur, and E. Braak, "Staging of brain pathology related to sporadic Parkinson's disease," Neurobiol. Aging, vol. 24, no. 2, pp. 197–211, 2003.

[6] L. Breiman, "Random forests," Mach. Learn., vol. 45, no. 1, pp. 5–32, 2001.

[7] T. Chen and C. Guestrin, "XGBoost: a scalable tree boosting system," in Proc. 22nd ACM SIGKDD Int. Conf. Knowledge Discovery Data Mining, 2016, pp. 785–794.

[8] J. Cohen, "Weighted kappa: nominal scale agreement with provision for scaled disagreement or partial credit," Psychol. Bull., vol. 70, no. 4, pp. 213–220, 1968.

[9] G. S. Collins, K. G. M. Moons, P. Dhiman, et al., "TRIPOD+AI statement: updated reporting guidelines for clinical prediction models that use regression or machine learning methods," BMJ, vol. 385, Art. no. e078378, 2024.

[10] C. Cortes and V. Vapnik, "Support-vector networks," Mach. Learn., vol. 20, no. 3, pp. 273–297, 1995.

[11] A. Diaz-Rincon et al., "Conformal prediction for Parkinson's disease medication needs," arXiv preprint, 2025.

[12] A. Bentivoglio et al., “Clinical and imaging characteristics of Parkinson’s disease with negative alpha-synuclein seed amplification assay,” Mov. Disord., 2026. doi: 10.1002/mds.70197.

[13] et al., “New diagnostic and staging framework applied to established PD in the BioFIND cohort,” npj Parkinsons Dis., vol. 11, 2025. doi: 10.1038/s41531-025-00992-3.

[14] A. J. Espay et al., “Clinical and biological characterization of patients with Parkinson’s disease with and without improvement on dopaminergic medications,” Lancet Neurol., 2025.

[15] A. J. Espay et al., “Refutation of the αSyn-SAA-based staging for Parkinson’s progression (Neuronal α-Synuclein Disease-Integrated Staging System [NSD-ISS]),” Mov. Disord., 2025. doi: 10.1002/mds.30269.

[16] P. Eusebi et al., “Diagnostic utility of [123I]FP-CIT SPECT in differentiating Parkinson’s disease from essential tremor: a meta-analysis,” Eur. J. Nucl. Med. Mol. Imaging, vol. 44, no. 3, pp. 495–514, 2017. doi: 10.1007/s00259-016-3597-9.

[17] B. Fischl, “FreeSurfer,” NeuroImage, vol. 62, no. 2, pp. 774–781, 2012. doi: 10.1016/j.neuroimage.2012.01.021.

[18] C. G. Goetz et al., “Movement Disorder Society-sponsored revision of the Unified Parkinson’s Disease Rating Scale (MDS-UPDRS): scale presentation and clinimetric testing results,” Mov. Disord., vol. 23, no. 15, pp. 2129–2170, 2008.

[19] L. Grinsztajn, E. Oyallon, and G. Varoquaux, “Why do tree-based models still outperform deep learning on tabular data?” in Proc. NeurIPS, 2022.

[20] K. Huang, Y. Jin, E. Candes, and J. Leskovec, “Uncertainty quantification over graph with conformalized graph neural networks,” in Proc. NeurIPS, 2023.

[21] G. Ke et al., “LightGBM: a highly efficient gradient boosting decision tree,” in Proc. NeurIPS, 2017.

[22] K. Marek et al., “The Parkinson’s Progression Markers Initiative (PPMI) — establishing a PD biomarker cohort,” Ann. Clin. Transl. Neurol., vol. 5, no. 12, pp. 1460–1477, 2018.

[23] V. Taquet, V. Blot, T. Morzadec, L. Lacombe, and N. Brunel, “MAPIE: an open-source library for distribution-free uncertainty quantification,” arXiv:2207.12274, 2022.

[24] B. Mollenhauer, C. J. Caspell-Garcia, C. T. Coffey, et al., “Longitudinal CSF biomarkers in patients with early Parkinson disease and healthy controls,” Neurology, vol. 89, no. 19, pp. 1959–1969, 2017. doi: 10.1212/WNL.0000000000004609.

[25] M. A. Nalls, C. Blauwendraat, C. L. Vallerga, et al., “Identification of novel risk loci, causal insights, and heritable risk for Parkinson’s disease: a meta-analysis of genome-wide association studies,” Lancet Neurol., vol. 18, no. 12, pp. 1091–1102, 2019. doi: 10.1016/S1474-4422(19)30320-5.

[26] R. B. Postuma et al., “MDS clinical diagnostic criteria for Parkinson’s disease,” Mov. Disord., vol. 30, no. 12, pp. 1591–1601, 2015.

[27] L. Prokhorenkova, G. Gusev, A. Vorobev, A. V. Dorogush, and A. Gulin, “CatBoost: unbiased boosting with categorical features,” in Proc. NeurIPS, 2018.

[28] E. Rissanen et al., “Machine learning applications for Parkinson’s disease: a systematic review,” NPJ Digit. Med., 2022.

[29] et al., “Reconsidering the clinical foundations of NSD-ISS in PD staging,” Mov. Disord., 2025. doi: 10.1002/mds.70061.

[30] M. J. Russo et al., “Application of the NSD-ISS biological staging framework in the BioFIND cohort,” npj Parkinson’s Dis., 2025.

[31] T. Simuni et al., “Reply to: Refutation of the αSyn-SAA-based staging for Parkinson’s progression,” Mov. Disord., 2025. doi: 10.1002/mds.30272.

[32] H. Schmitz-Steinkrüger, C. Lange, I. Apostolova, F. L. Mathies, L. Frings, S. Klutmann, S. Hellwig, P. T. Meyer, and R. Buchert, “Impact of age and sex correction on the diagnostic performance of dopamine transporter SPECT,” Eur. J. Nucl. Med. Mol. Imaging, vol. 48, no. 5, pp. 1445–1459, May 2021. doi: 10.1007/s00259-020-05085-2.

[33] J. P. Seibyl, “DaT-SPECT in Parkinson’s disease: clinical utility and limitations,” J. Nucl. Med., 2018.

[34] G. Shafer and V. Vovk, “A tutorial on conformal prediction,” J. Mach. Learn. Res., vol. 9, pp. 371–421, 2008.

[35] R. Shwartz-Ziv and A. Armon, “Tabular data: Deep learning is not all you need,” Information Fusion, vol. 81, pp. 84–90, 2022.

[36] T. Simuni et al., “A biological definition of neuronal alpha-synuclein disease: towards an integrated staging system for research,” Lancet Neurol., vol. 23, no. 2, pp. 178–190, Feb. 2024.

[37] T. Simuni et al., “Validation of the NSD-ISS in PPMI, PASADENA, and SPARK,” npj Parkinson’s Dis., 2024.

[38] T. Simuni et al., “Longitudinal clinical and biomarker characteristics of non-manifesting carriers and participants with Lewy body spectrum disease in the PPMI study using the NSD-ISS: A prospective cohort study,” Lancet Neurol., vol. 24, no. 2, pp. 127–142, Feb. 2025.

[39] P. Veličković, G. Cucurull, A. Casanova, A. Romero, P. Liò, and Y. Bengio, “Graph attention networks,” in Proc. Int. Conf. Learn. Representations (ICLR), 2018.

[40] A. Varrone et al., “European multicentre database of healthy controls for [123I]FP-CIT SPECT (ENC-DAT): age-related effects, gender differences and evaluation of different methods of analysis,” Eur. J. Nucl. Med. Mol. Imaging, vol. 40, no. 2, pp. 213–227, 2013. doi: 10.1007/s00259-012-2276-8.

[41] V. Vovk, A. Gammerman, and G. Shafer, Algorithmic Learning in a Random World, 2nd ed. Springer, 2022.

[42] N. Wakasugi, H. Takano, M. Abe et al., “Harmonizing multisite data with the ComBat method for enhanced PD diagnosis via DAT-SPECT,” Front. Neurol., vol. 15, Art. no. 1306546, 2024. doi: 10.3389/fneur.2024.1306546.

[43] H. Zou and T. Hastie, “Regularization and variable selection via the elastic net,” J. R. Stat. Soc. B, vol. 67, no. 2, pp. 301–320, 2005.